Differential Geometry 6

1) Warm-up: base D^2 (contractible) $\Rightarrow$ everything is trivial, parallel transport can be flat

Let

$$M = D^2 \subset \mathbb{R}^2.$$

(a) Trivial vector bundle and sections. Take

$$E = D^2 \times \mathbb{R}^2, \qquad \pi(x, v) = x.$$

A section is just a map

$$s(x) = (x, V(x)),$$

i.e. a vector field

$$V : D^2 \to \mathbb{R}^2.$$

(b) Frame bundle (principal $\mathrm{GL}(2)$). The frame bundle

$$F(E) \to D^2$$

is also trivial:

$$F(E) \cong D^2 \times \mathrm{GL}(2, \mathbb{R}),$$

and a (global) section is a global frame—e.g. the constant frame

$$(e_1, e_2).$$

(c) A flat connection and parallel transport. Take the flat connection (think: the usual derivative in $\mathbb{R}^2$). Its curvature is

$$0,$$

so parallel transport along any loop is the identity (path-independent on simply-connected sets).

Moral. Over

$$D^2$$

you can always pick global sections/frames and choose flat connections.

2) Base S^2 (curved) $\Rightarrow$ tangent bundle is nontrivial, Levi–Civita transport has nontrivial holonomy

Let

$$M = S^2 \subset \mathbb{R}^3$$

with the round metric (unit sphere).

(a) Tangent bundle and sections. Take the tangent bundle

$$\pi : TS^2 \to S^2, \qquad \pi(p, v) = p,$$

whose fiber at p is the tangent plane

$$(TS^2)_p = T_pS^2.$$

A section

$$s : S^2 \to TS^2$$

is a tangent vector field:

$$s(p) = (p,\, V(p)), \qquad V(p) \in T_pS^2.$$

(There is no nowhere-vanishing global tangent field on S^2, hence no global frame of TS^2.)

(b) Orthonormal frame bundle (principal SO(2)). Define the oriented orthonormal frame bundle

$$P := F^+_{\mathrm{SO}}(S^2) = \big\{(p, e_1, e_2) : p \in S^2,\ (e_1, e_2) \text{ oriented orthonormal basis of } T_pS^2\big\}.$$

The projection

$$\pi : P \to S^2, \qquad \pi(p, e_1, e_2) = p$$

makes P a principal SO(2)-bundle, with right action given by rotating the frame:

$$(p, e_1, e_2) \cdot R = (p,\, e_1', e_2'), \qquad R \in \mathrm{SO}(2).$$

(c) Levi–Civita connection as a principal connection. The Levi–Civita connection on TS^2 induces a principal connection on P: equivalently, a choice of horizontal subspaces

$$T_uP = H_u \oplus V_u \qquad (u \in P),$$

where V_u are the vertical (fiber) directions.

In spherical coordinates (θ, ϕ) on S^2, consider the orthonormal tangent frame

$$e_\theta = \partial_\theta, \qquad e_\phi = \frac{1}{\sin\theta}\,\partial_\phi.$$

Then the Levi–Civita connection can be encoded by a single $\mathfrak{so}(2)$-valued 1-form

$$\omega = \cos\theta\, d\phi$$

(up to a sign convention), describing how the frame rotates along motion on the sphere.

(d) Parallel transport and holonomy ("area gives rotation"). Given a curve

$$\gamma : [0,1] \to S^2,$$

parallel transport for Levi–Civita means: a vector field $V(t) \in T_{\gamma(t)}S^2$ satisfying

$$\nabla_{\dot\gamma(t)} V(t) = 0.$$

For a closed loop $\gamma = \partial\Omega$, parallel transport typically returns a vector rotated by a holonomy angle. On the unit sphere (Gaussian curvature $K \equiv 1$), for sufficiently nice loops,

$$\text{holonomy angle} = \int_\Omega K\, dA = \mathrm{Area}(\Omega).$$

Concrete triangle example. Let Ω be the spherical triangle with vertices: north pole N and two points A, B on the equator whose longitudes differ by α, with edges: meridian $N \to A$, equator $A \to B$, meridian $B \to N$. Then the enclosed area is

$$\mathrm{Area}(\Omega) = \alpha$$

(unit sphere), hence parallel transport around the triangle rotates tangent vectors by angle

$$\alpha.$$

Moral. Over

$$S^2$$

the tangent bundle and frame bundle are globally nontrivial, and Levi–Civita parallel transport around loops produces nontrivial holonomy determined by curvature (on the unit sphere: by area).

3) Hopf fibration: a nontrivial principal S^1-bundle over S^2 with an explicit connection

Let

$$S^3 = \{(z_1, z_2) \in \mathbb{C}^2 : \ |z_1|^2 + |z_2|^2 = 1\}.$$

(a) Principal S^1-bundle structure. Let S^1 act on S^3 by

$$e^{i\psi} \cdot (z_1, z_2) = (e^{i\psi} z_1,\ e^{i\psi} z_2).$$

The quotient space is

$$S^3/S^1 \cong \mathbb{C}P^1 \cong S^2,$$

so we get a principal bundle

$$S^1 \hookrightarrow S^3 \xrightarrow{p} S^2,$$

called the Hopf fibration.

A concrete Hopf map into $\mathbb{R}^3$ (identifying $S^2 \subset \mathbb{R}^3$) is

$$p(z_1, z_2) = \Big(2\Re(z_1\overline{z_2}),\ 2\Im(z_1\overline{z_2}),\ |z_1|^2 - |z_2|^2\Big),$$

which indeed lies on S^2.

(b) Local sections (local triviality). Cover S^2 by two charts U_+ (remove the south pole) and U_- (remove the north pole). Over each chart, the bundle is trivial:

$$p^{-1}(U_\pm) \cong U_\pm \times S^1,$$

so one can choose local sections

$$s_\pm : U_\pm \to S^3.$$

On the overlap $U_+ \cap U_-$ the sections differ by a transition function

$$g_{+-} : U_+ \cap U_- \to S^1,$$

which encodes the global nontriviality of the bundle.

(c) An explicit connection 1-form. There is a standard S^1-connection on the Hopf bundle. In Euler-angle coordinates (θ, ϕ, ψ) on S^3 adapted to the fibration, one can take

$$\omega = d\psi + \cos\theta \, d\phi.$$

Here:

- ψ is the fiber coordinate (along S^1),
- (θ, ϕ) are coordinates on the base S^2.

Equivalently (in complex coordinates), one often writes the same connection as

$$\omega = \Im\big(\overline{z_1}\, dz_1 + \overline{z_2}\, dz_2\big),$$

restricted to S^3.

(d) Curvature and holonomy. The curvature 2-form is

$$\Omega = d\omega,$$

and (up to a conventional normalization factor) it pulls back the area form on S^2. Hence the holonomy around a loop $\gamma = \partial\Omega$ in the base is governed by the enclosed area: transport around γ produces a phase in S^1 determined by integrating curvature over the spanned region.

Moral. The Hopf fibration is a clean, explicit example of: a nontrivial principal bundle over S^2, local sections but no global section compatible with trivialization, and a canonical connection whose curvature detects area (and thus yields nontrivial holonomy).